clustering for your applications so that you can run multiple instances of your worker service and share jobs among them.

All the background tasks you've seen so far in this chapter repeat a task on an interval indefinitely, from the moment the application starts. However, sometimes you want more control of this timing. Maybe you always want to run the application at 15 minutes past each hour. Or maybe you want to run a task only on the second Tuesday of the month at 3 a.m. Additionally, maybe you want to run multiple instances of your application for redundancy but ensure that only one of the services runs a task at any time.

It would certainly be possible to build all this extra functionality into your app yourself, but excellent libraries already provide all this functionality for you. Two of the most well known in the .NET space are Hangfire (https://www.hangfire.io) and Quartz.NET (https://www.quartz-scheduler.net).

Hangfire is an open-source library that also has a Pro subscription option. One of its most popular features is a dashboard UI that shows the state of all your running jobs, each task's history, and any errors that have occurred.

Quartz.NET is completely open-source and essentially offers a beefed-up version of the `BackgroundService` functionality. It has extensive scheduling functionality, as well as support for running in a clustered environment, where multiple instances of your application coordinate to distribute the jobs among themselves.

NOTE Quartz.NET is based on a similar Java library called Quartz Scheduler. When looking for information on Quartz.NET, be sure you're looking at the correct Quartz!

Quartz.NET is based on four main concepts: